

# State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts  
Key insights from JetBrains' survey of 23,262 developers worldwide

23,262 developers

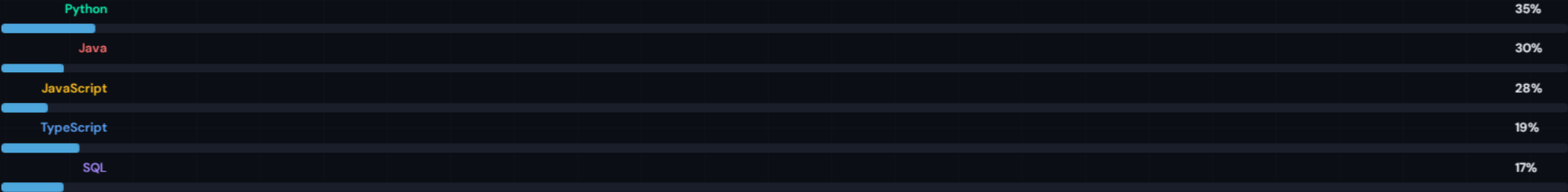
2017–2024 trend data

20+ languages tracked

# Python Closes the Gap: 8 Years of Language Usage Shifts

JavaScript **peak** → **decline** 70% → 61%Python **steady rise** 32% → 57%TypeScript **3x growth** 12% → 37%PHP **-13pp** 30% → 17%

# Python Is Now the #1 Primary Language at 35%



KEY INSIGHT

Adoption intent stays below 6% for all languages — developers stick with what they know.

LEGEND

- Primary language
- Planning to adopt

# TypeScript and Rust Top Developer Satisfaction, **Nearly Tied** at 199 vs 198



## PATTERN

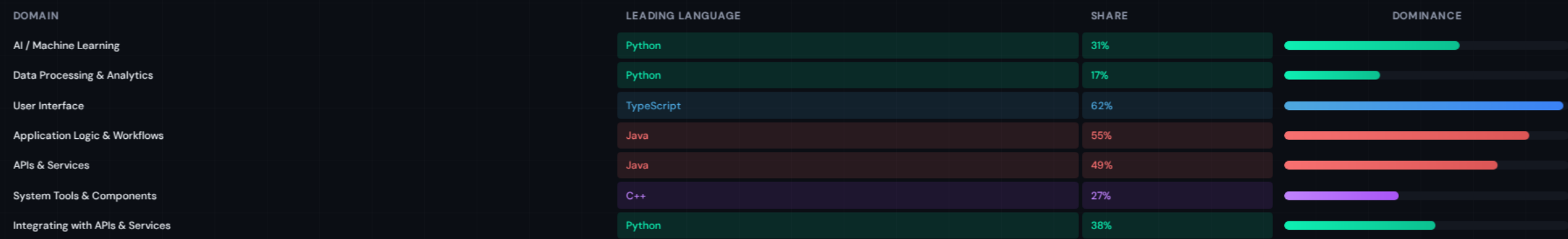
Type-safe languages with modern tooling dominate the top spots.

## TOP 2

**199** vs **198**

TypeScript vs Rust

# Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript



Python (3 domains) TypeScript (1 domain) Java (2 domains) C++ (1 domain)

## What This Means for Your **Technology Strategy** in 2025

- 01** **AI / ML** Invest in Python for AI/ML teams — it is the undisputed leader at 31% share in AI and 28% in data science.
- 02** **Frontend** TypeScript is the future of frontend and full-stack — 62% UI share, #1 in developer satisfaction at 199.
- 03** **Infra** Rust and Go are the growth plays for infrastructure and systems — both designed for performance and compiler safety.
- 04** **Shrinking** PHP and Ruby hiring pools are shrinking — PHP dropped from 30% to 17%, Ruby from 10% to 4% since 2017.
- 05** **Retention** Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy.